

Anatomy of a Single Intervention Step

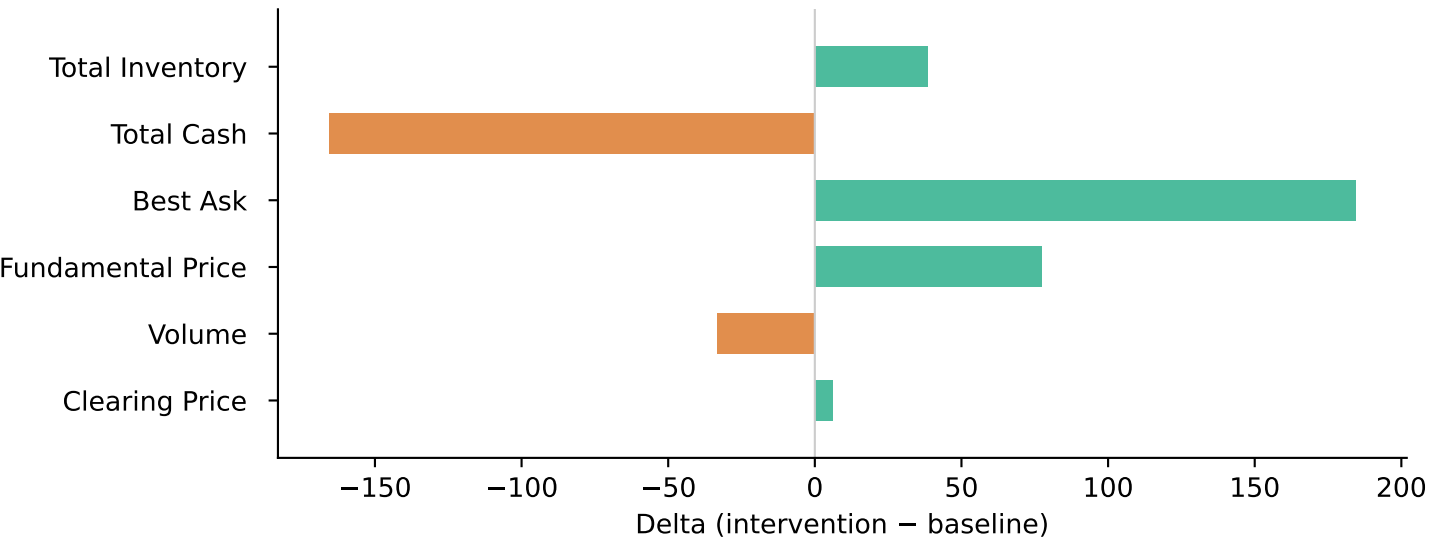
INTERVENTION (Step 1)

Type: EVENT — Supply Disruption (magnitude=3.0)

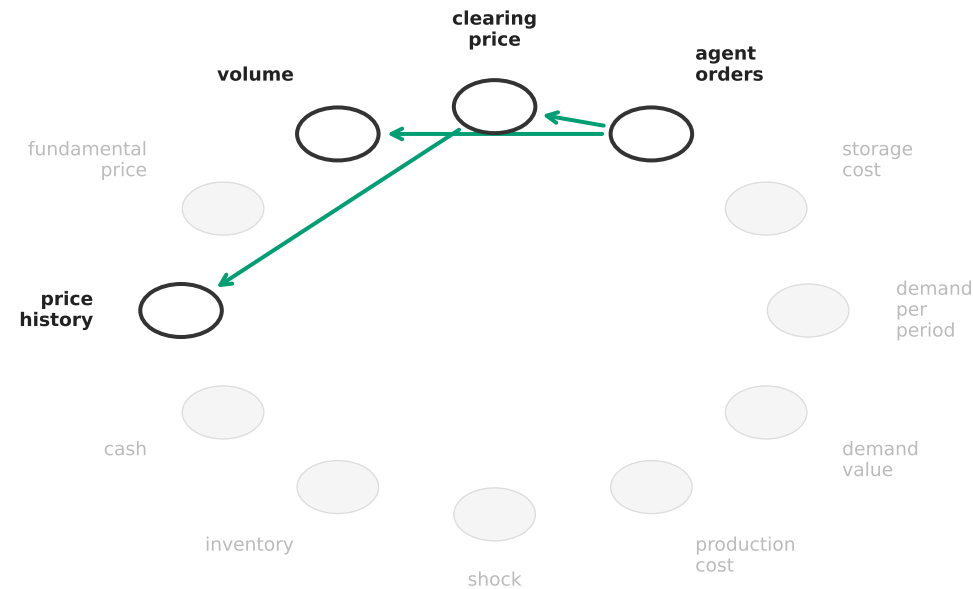
Purpose: Test shock → production\_cost → fundamental\_price pathway

Mechanism: Clamp shock variable, run 3 periods, compare vs. no-intervention baseline

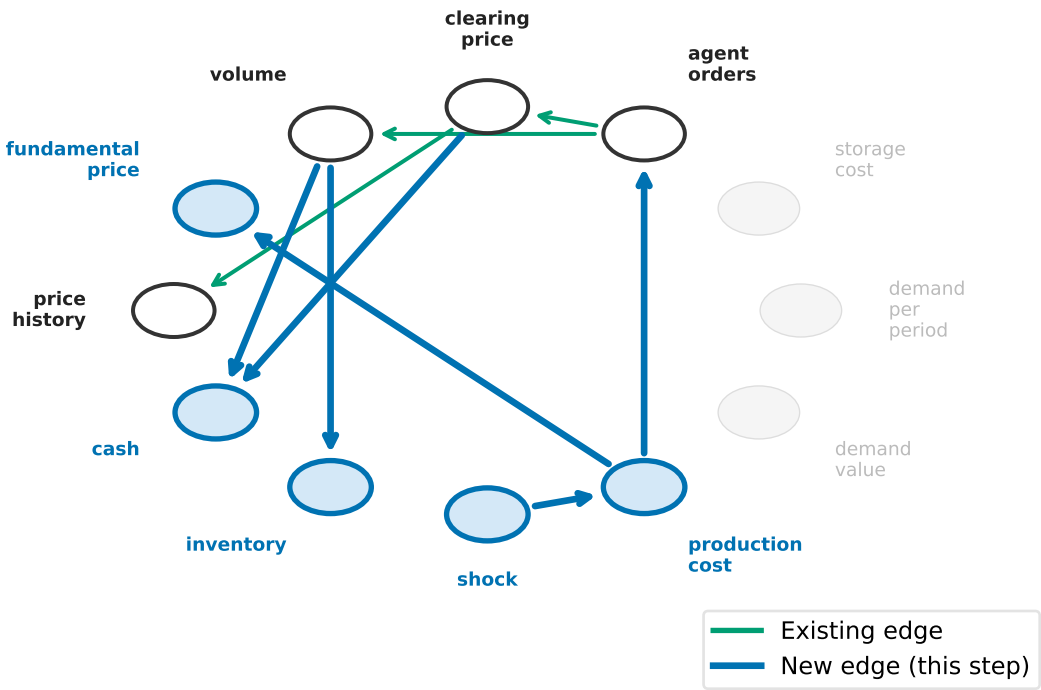
Observed Effects



Graph BEFORE Intervention (3 edges)



Graph AFTER Intervention (9 edges, +6 new)



MODEL REASONING (Gemini 3.1 Pro)

Supply shock caused massive increases in fundamental\_price (+77) and best\_ask (+184), while bid prices remained unchanged. This confirms the shock affects supply-side parameters (production\_cost), driving up fundamental\_price and ask prices. Volume drop (−33) and cash/inventory changes are downstream effects.

Key uncertainties:

- Does fundamental\_price directly influence agent\_orders?
- Does price\_history influence agent\_orders (technical trading)?